

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our recent analytical initiative has successfully developed a highly effective predictive model, capable of accurately forecasting outcomes for over 83% of cases. This strong performance provides a robust and reliable foundation for deriving strategic insights and informing critical business decisions. The analysis clearly identifies three primary factors that significantly influence these outcomes: gender (specifically male), the price paid for the service or product, and the individual's age. Gender appears to be the most critical determinant of predictions, with the associated price and the age of the individual also playing substantial roles in shaping the results. Given the high reliability of our predictive model and the clarity of these key drivers, we strongly recommend close and regular monitoring of these variables. Any shifts or changes in these factors – particularly gender demographics, pricing strategies, or age profiles – are expected to have the most significant impact on the business outcomes we are forecasting. These insights offer valuable guidance for optimizing strategic planning, targeting specific initiatives, and allocating resources more effectively.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

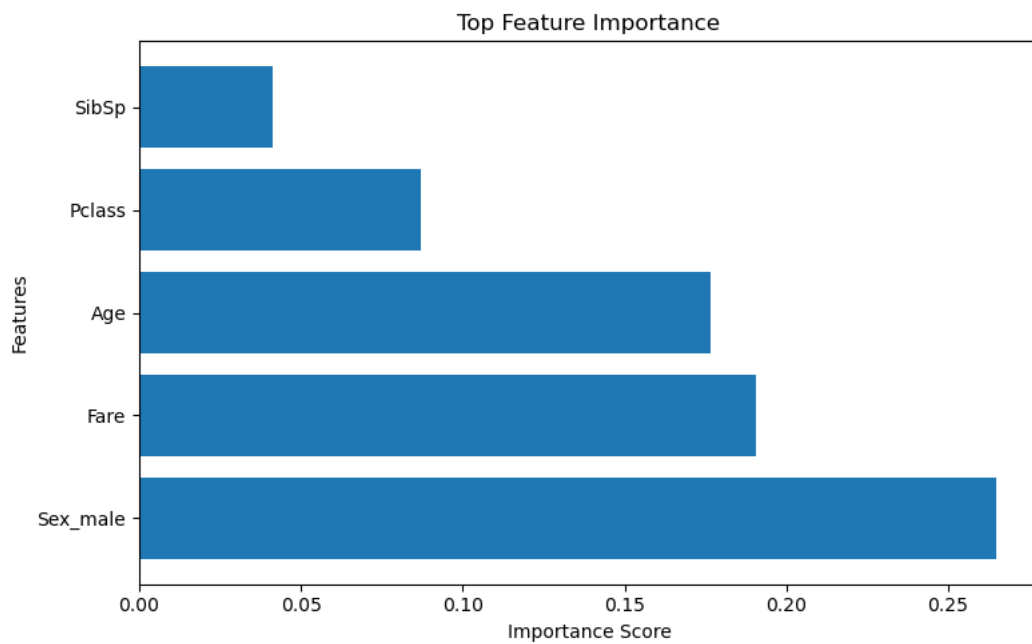
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

